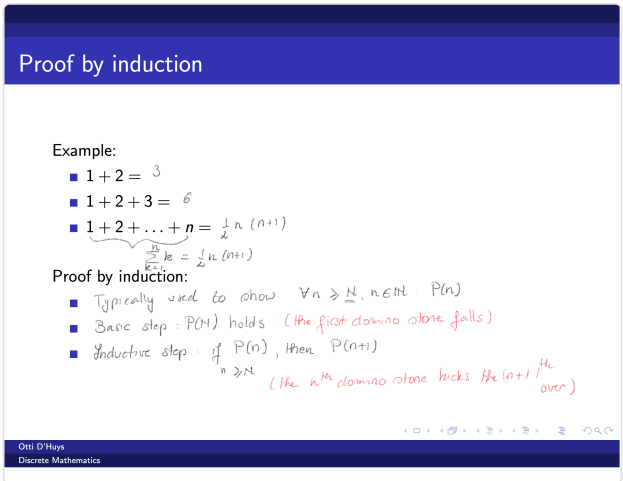
BCS1130-4 Proof by induction

Lecture slides



Lecture notes

Can you calculate the sum 1+2+3 + ….+ (n-1) + n (with n a natural number)

- $\sum_{i=1}^{n}(i) = 1 + 2 + \cdots + n = \frac{n}{2}(n+1)$
  - We can demonstrate this with a few examples
  - Or we can demonstrate it through mathematical induction

Mathematical induction theory

Mathematical induction

Proof technique for statements of the form $\forall n \geq N, n \in Z$ (set of integers from a starting point on, e.g. natural numbers)

- Two steps in the proof by induction:
  - **==Base case ==**(proof of $P(n)$) $\rightarrow$ the statement holds for the first number
  - **Induction step**: proof of $\forall n \geq N(P(n) \rightarrow P(n+1))$
- Process of induction
  - $P(n)$ is true $\rightarrow$ base case
  - $P(n) \rightarrow P(n+1) \forall n \rightarrow$ (induction step)
  - $P(n), P(n) \rightarrow P(n+1) \implies P(n+1)$

Induction example

Induction example

$$\forall n \in N \left(\sum_{k+1}^{n} k = \frac{n(n+1)}{2} \right)$$

Base case: Proof of P(1)

- $\sum_{k=1}^{1} = 1 = \frac{1(1+1)}{2} = 1 \rightarrow$ This holds

Induction step: Proof of $\forall m \geq 1, p(m) \rightarrow P(m+1)$

- $p(m) : \sum_{k=1}^{m} k = \frac{m(m+1)}{2}$ (assumption)
- $p(m+1) : \sum_{k=1}^{m+1} k = \frac{(m+1)(m+2)}{2}$ (to prove)

- $\sum_{k=1}^{m+1} k = (1 + 2 + \ldots. + m + (m+1))$
  - we see that $(1 + 2 + \ldots + m) = p(m) : \sum_{k=1}^{m} k = \frac{m(m+1)}{2}$
- We use our assumption
  - $\sum_{k=1}^{m+1} k = (1 + 2 + \ldots. + m + (m+1)) = \frac{m(m+1)}{2} + (m+1)$
  - $= (m+1)\left(\frac{m}{2} + 1\right) = (m+1)\left(\frac{m+2}{2}\right)$

Induction example 2

Induction example 2

$\forall n \in N : 4^n - 1$ is divisible by 3

Base case: Proof of $P(1)$

- $P(1) = 4^1 - 1 = 3 \rightarrow$ 3 is divisible by 3$

Induction step: Proof of $P(x) \rightarrow P(x+1) \forall x \in N$

- $P(x) : 4^x - 1$ is divisible by 3 (assumption / hypothesis)
- $P(x+1) : 4^{x+1} - 1$ is divisible by 3 (to prove)

Always use your assumption hypothesis!

- $P(x+1) = 4^{x+1} - 1 = 4 \cdot 4^x - 1 = 3 \cdot 4^x + 4^x - 1$
  - $4^x - 1$ is divisible by 3 by assumption $(3k) \rightarrow 4^x - 1 = 3k$
  - $3 \cdot 4^x$ is also a multiple of 3
- $P(x+1) = 3 \cdot 4^x + 4^x - 1 = 3 \cdot 4^x + 3k = 3(4^x + k) \rightarrow$ multiple of 3

Induction example 3

- [ ] **Ask for notes on this example cause it was confusing as fuck**

Induction example 3

A country has two types of coins: 3 cents and 7 cents. Prove that you can make all sums of money starting from 12 cents (up to 1 cent accuracy) from these coins.

Explaining the example

$2 \cdot 7 = 5 \cdot 3$

Induction proof 3.loom

Negation: max 1x7, max 1x3 → cannot add up to 12 or more

1. If we have 2x3 (at least), we can go 1 cent up
2. If we have 2x7 (at least), we can go 1 cent up
3. We always have to have either (or both) 2x7, or 2x3

Proof by induction

- **Base case:** $P(12)$ holds $\rightarrow 12 = 3 + 3 + 3 + 3$
- **Inductive step:** $P(q) \rightarrow P(q+1)$

1. q has at least 2 coins of 7 or 2 coins of 3
    - r: at least 2 coins of 7
    - s: at least 2 coins of 3
    - $\neg(r \vee s) = \neg r \wedge \neg s$ is impossible, as q ≥ 1
2. if r is true, then we can replace 3x7 by 5x3
3. if s is true, then we can replace 2x3 by 7

- → in both cases, we can make q + 1

Induction example 4

⚠ **An example like this won't show up in the exam**

Induction example 4 (Disproving with induction)

$$\forall x \in N, \exists y \in N : x^y < y^x$$

Counterexample: x = 3
To prove: $\exists x \forall y \in N : 3^y \geq y^3$ (negation of the statement, with x=3)

Base case:
$y = 1 \rightarrow 3^1 \geq 1^3 \rightarrow$ this is true
$y = 2 \rightarrow 3^2 \geq 2^3 \rightarrow$ true
$y = 3 \rightarrow 3^3 = 3^3 \rightarrow$ true

Induction step: if $3^k \geq k^3 \implies (3^{k+1} \geq (k+1)^3)$ for all $k \geq 3$

$$\forall x \in \mathbb{N}, \exists y \in \mathbb{N} : x^y < y^x$$

Counter example : $x = 3$

To prove : $\forall y \in \mathbb{N} : 3^y \geq y^3$

$y = 1 \rightarrow 3^1 \geq 1^3$ ✓

$y = 2 \rightarrow 3^2 \geq 2^3$ ✓

$y = 3 \rightarrow 3^3 = 3^3$ ✓ $\rightarrow$ base case.

- Induction step : if $(\underline{3^k \geq k^3}) \rightarrow (3^{k+1} \geq (k+1)^3)$ for all $k \geq 3$

$$\begin{aligned}(k+1)^3 &= k^3 + 3k^2 + 3k + 1 \\ &\leq 3^k + (3k^2 + 3k + 1) \\ &\leq 3^k + k^3 + 3k + 1 \\ &\leq 3^k + 3^k + 3^k \\ &\leq 3^{k+1}\end{aligned}$$ ✓

$3k + 1 \leq k^3$

$\Leftrightarrow 1 \leq k^3 - 3k$

$\Leftrightarrow 1 \leq \underbrace{k}_{\leq 3}\underbrace{(k^2 - 3)}_{\geq 6}$

true!